

Rebuttal for CoRL Submission #2468

We thank the reviewers and Area Chair for their constructive questions. The additional analyses strengthen the paper’s central conclusion that an online, tactile-supervised future-contact prior provides information beyond nominal task progress and helps a diffusion policy overcome pre-contact stall. They also identify the mechanisms responsible for this gain, motivate two cleaner implementation refinements—fixed bounded loss weighting and tactile-only masking—and make the current evaluation scope explicit.

Q1: Contact anticipation versus task progress. We thank the reviewer for pointing us to Hierarchical Diffusion Policy (Wang et al., T-RO 2025) and CompILE (Kipf et al., ICML 2019); we will add these references in the camera-ready version. Phase/progress conditions are useful time-indexed descriptions of nominal task order, whereas VPCE is a state-conditioned, tactile-supervised 15-step contact horizon predicted from current wrist RGB and recent robot motion. It conditions the same diffusion action generator continuously, rather than switching discrete phases or sub-policies. On held-out trajectories, VPCE reaches AUC/F1 of .963/.759, .885/.705, and .946/.735 on SWIPE/WIPE/INSERT, compared with .853/.597, .743/.406, and .808/.539 for the stronger progress-only predictor (the phase-only baseline obtains .670/.000, .666/.000, and .684/.000). Thus nominal progress explains part of contact timing, but the current visual state and motion provide additional contact-specific information. We retain the more detailed predictor comparison below and do not claim universal zero-shot transfer: target-domain tactile adaptation remains part of the current method.

TABLE I
CONTACT PREDICTION AUC/F1 USING PHASE, PROGRESS, OR VPCE.

Task	Phase-4	Progress-20	VPCE
SWIPE	.670/.000	.853/.597	.963/.759
WIPE	.666/.000	.743/.406	.885/.705
INSERT	.684/.000	.808/.539	.946/.735

Progress captures some timing, but vision, motion, and tactile-derived contact supervision add information beyond normalized trajectory time. This distinction also explains the synergy with tactile feedback. VPCE supplies information before touch becomes informative—*when* to commit to contact—whereas tactile observations regulate *how* to execute after contact. The submitted 2×2 comparison separates these roles: on SWIPE, the contact-prior-only policy reaches 85% versus 10% for tactile-only DP, and the full method reaches 90%; on WIPE/INSERT, the full method reaches 70/45%, compared with 50/30% for prior-only and 60/15% for tactile-only policies. Thus contact anticipation and tactile feedback are complementary rather than competing representations. The zero-shot results further delineate the current transfer boundary: source-teacher AUC/F1 is .500/.095 on board and .662/.666

on card, while target adaptation raises board performance to .897/.750. The current target-domain teacher uses 60 demonstrations with tactile recordings per downstream task; contact targets are generated automatically from the recorded tactile signals, without manual frame-level annotation. The current evidence therefore supports a contact-specific signal beyond progress, with target adaptation improving transfer across new visual/contact domains.

Q2: Loss-weight optimization. We thank the reviewer for identifying this optimization ambiguity. The original design intuition was to emphasize sparse but control-critical approach-to-contact transitions. With uniform denoising loss, abundant and relatively easy free-space samples can dominate optimization, even though small action errors near contact determine whether engagement succeeds. We therefore used the VPCE mean probability c to increase the gradient contribution of contact-imminent samples. The learnable γ was intended to adapt the global strength of this emphasis across tasks, while the residual δ was intended to compensate for the fact that contact probability and action difficulty are not identical; $\text{softplus}(\gamma)$ kept the base gain non-negative. We agree, however, that optimizing these parameters through the same weighted objective creates a downward incentive. Checkpoint inspection confirms that the resulting weight is approximately 1.000–1.005: the lower bound prevents harmful down-weighting, but largely reduces the term to ordinary DP loss. We retain the original emphasis intuition while removing this ambiguity with the fixed bounded schedule $w(c) = \text{clip}(1 + c, 1, 2)$, which smoothly maps free-space samples to weight 1 and high-contact-probability samples to at most 2, without a learned collapse direction. With $\alpha=1$, this fixed schedule obtains 8/10 on SWIPE and 7/10 on WIPE. We treat these small- N evaluations as a functionality check rather than evidence that loss weighting alone improves success. Together with the submitted no-loss-weighting result (8/10 on SWIPE), they show that the central contact-anticipation effect does not depend on the original learned weighting term, while the fixed schedule retains the intended emphasis on contact-critical samples without an optimization shortcut.

Q3: Masking and free-space motion. We thank the reviewer for highlighting that Eq. (9) conflated distinct low-dimensional signals. The original post-encoder block included proprioception, gripper width, and tactile features, although the intended inductive bias was only to suppress uninformative pre-contact touch. We therefore ran a controlled mask-scope retraining that changes only this routing: the revision always retains proprioception and gripper width and masks only tactile/contact features. It obtains 13/20 on SWIPE. Qualitatively, preserving proprioception produces more coherent free-space approach motion and avoids the endpoint-seeking/repositioning behavior observed with the whole-lowdim mask, while contact-adaptive tactile masking

still suppresses pre-contact sensor variation. Action-chunk prediction and receding-horizon execution provide additional temporal coherence. This result supports masking as tactile routing, rather than as suppression of robot state.

Q4: Gating and component ablations. We thank the Area Chair and Reviewer Fcwd for emphasizing that the component ablations should be visible in the main paper rather than only in the appendix; this will make the evidence for each design choice easier to assess. The submitted ablation showed that removing horizon conditioning or masking reduced SWIPE to 2/10 and 0/10, whereas removing gating gave 9/10 and removing loss weighting gave 8/10. Together with the new no-gating (8/20) and bounded-gate (7/20) retrains, these experiments identify a clear division of labor: future-contact horizon conditioning and contact-adaptive masking are the main drivers of task success; loss weighting is a secondary training emphasis; and gating is an auxiliary, task-dependent modulation rather than a required source of the contact-anticipation gain. We will move the compact component-ablation table and this interpretation into the main text in the revision, and retain the full behavioral analysis in the appendix.

Q5: Operator-held evaluation protocol. We thank the reviewer for highlighting that the supplementary video did not make this protocol and its rationale sufficiently clear. The reader and foam were held only to maintain a predefined height and orientation in the robot’s reachable workspace; the holder did not track the end effector, adjust the object online, or assist alignment after a rollout began. The same initial placement, success criterion, and 60-s timeout were used across methods. We expanded the original VPCE-DP evaluation to 30 trials per task, obtaining 28/30 on SWIPE (93.3%, Wilson 95% CI [78.7, 98.2]%), 22/30 on WIPE (73.3%, [55.6, 85.8]%), and 18/30 on INSERT (60.0%, [42.3, 75.4]%). These additional rollouts are consistent with the submitted results and reduce sampling uncertainty.

The standardized height also exposes an important scope condition. VPCE predicts only a 15-step future-contact horizon aligned with the policy horizon; it is not a long-range approach planner. If the interaction object begins far outside this anticipation range, predicted contact probability can remain low and the policy may still stall before reaching the pre-contact region. We agree that operator compliance may ease contact and will state this limitation explicitly. A natural extension is an external-view camera or coarse approach controller that first brings the end effector near the interaction region, after which the wrist-based VPCE regulates imminent contact. We will also investigate a height-matched rigid or force-limited fixture for standardized evaluation.

Q6: Contact-label calibration and contact direction. We thank the reviewer for raising this important extension. In the current setup, two planar fingertip tactile sensors and the gripper/task geometry yield a repeatable dominant interaction direction. We therefore derive the binary contact event from the temporal change of a base-frame z -force proxy; the same threshold of 0.095 is reused for most tasks. For the few domains with a different sensor scale, the threshold is selected

once from the training-demonstration force-change distribution and then held fixed. All frame labels are generated automatically from the tactile stream; no manual frame-level contact annotation is required. This is sufficient for the present tasks because VPCE predicts whether contact is imminent, rather than regressing the full contact wrench. We agree that dexterous hands, curved tactile surfaces, and contacts with changing normals require direction-agnostic or local-frame labels. A direct extension is to threshold the resultant change $\|\Delta \mathbf{F}\|_2$, or combine local normal and shear changes with hysteresis or learned change-point detection. Importantly, this replaces only the tactile-to-contact label adapter: the VPCE predictor, future-contact horizon, and policy-conditioning framework remain unchanged. We will add a threshold sensitivity analysis and investigate calibration-light multi-axis labeling in the revision and future work.

Scope. Target adaptation defines the current transfer boundary. The three tasks provide complementary evidence across precise engagement, sustained surface contact, and insertion. SWIPE shows the largest gain, while WIPE and INSERT test whether the same prior remains useful when post-contact regulation becomes increasingly important. The current 15-step predictor addresses short-horizon contact anticipation, not arbitrary long-range approach. We report Wilson intervals and avoid extrapolating beyond the evaluated task set.